

Two-Way Anova and Regression

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Introduction

In this report we are going to apply a simple two-way ANOVA for a data set consisting of 3 rows for the three different values of factor A, 3 columns for the 3 different values of parameter B and 4 different responses in each of the 9 created cells for a total of 36 response values.

The Data

The dataset was given in class and is the following:

Table 1: Data table

A	1	2	3
1	130, 74, 155, 180	34, 80, 40, 75	20, 82, 70, 58
2	150, 159, 188, 126	136, 106, 122, 115	25, 58, 70, 45
3	138, 168, 110, 160	174, 150, 120, 139	96, 82, 104, 60

Two-Way ANOVA and Results

We then apply the two-way ANOVA and print the summary:

```
model<-aov(Response~A*B, data=df)
summary(model)
```

```
          Df Sum Sq Mean Sq F value    Pr(>F)
A           2  10684     5342   7.911 0.00198 **
B           2   39119    19559  28.968 1.91e-07 ***
A:B          4    9614     2403   3.560 0.01861 *
Residuals   27  18231        675
---
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Parameter Estimation and Model

```
Total_Mean <- mean(df$Response)
```

First we calculate the Total Mean=105.53.

```
A_mean= df %>% group_by(A) %>% summarize(Amean=mean(Response), .groups="drop")
B_mean= df %>% group_by(B) %>% summarize(Bmean=mean(Response), .groups="drop")
AB_mean= df %>% group_by(A,B) %>% summarize(ABmean=mean(Response), .groups="drop")
```

We then calculate the $Y_{i..}$, $Y_{.j.}$, $Y_{ij.}$.

Using these we calculate the model parameters:

```
a_hat_table<-data.frame(a_hat = a_hat)
kable(a_hat_table, caption = "Estimates of  $\hat{a}_i$ ")
```

Table 2: Estimates of $\hat{a}_i$

<u>a_hat</u>
-22.36
2.81
<u>19.56</u>

```
b_hat_table<-data.frame(b_hat = b_hat)
kable(b_hat_table, caption = "Estimates of  $\hat{b}_i$ ")
```

Table 3: Estimates of $\hat{b}_i$

<u>b_hat</u>
39.31
2.06
<u>-41.36</u>

```
ab_hat_table<-data.frame(ab_hat = ab_hat)
kable(ab_hat_table, caption = "Estimates of  $\hat{ab}_i$ ")
```

Table 4: Estimates of $\hat{ab}_i$

ab_hat
12.28
-27.97
15.69
8.11
9.36
-17.47
-20.39
18.61
1.78

Lastly the residuals:

Table 5: $\hat{e}_{ijk}$

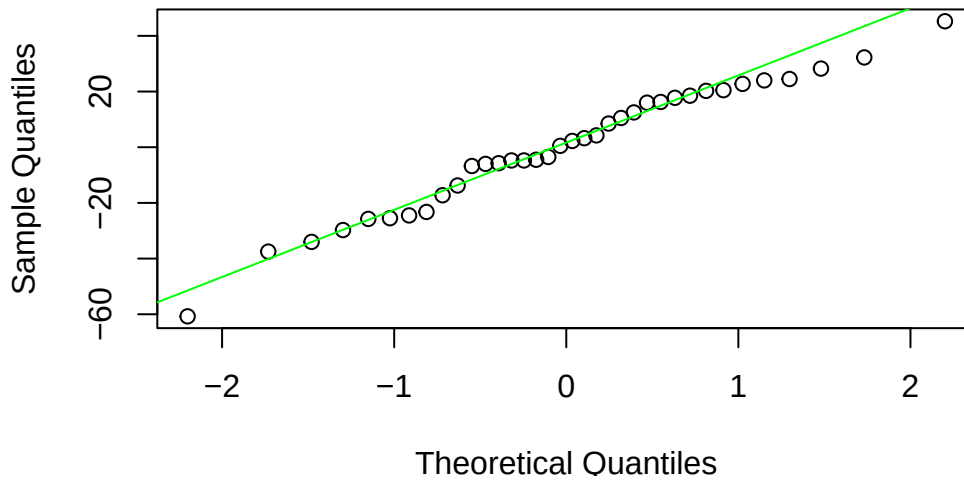
A	1	2	3
1	-4.75, -60.75, 20.25, 45.25	-23.25, 22.75, -17.25, 17.75	-37.5, 24.5, 12.5, 0.5
2	-5.75, 3.25, 32.25, -29.75	16.25, -13.75, 2.25, -4.75	-24.5, 8.5, 20.5, -4.5
3	-6, 24, -34, 16	28.25, 4.25, -25.75, -6.75	10.5, -3.5, 18.5, -25.5

Normal Probability Plot(Q-Q)

We will now draw a Normal Probability Plot(Q-Q) to check if our residuals follow a normal distribution:

```
qqnorm(residuals(model))
qqline(residuals(model), col="green")
```

Normal Q-Q Plot



The residuals seem to follow a normal distribution, thus reinforcing the ANOVA assumptions.

Regression with orthogonal contrasts

We are now going to repeat this results with a different approach, through a regression with orthogonal contrasts, specifically with the contrast vectors $A1, B1 = (-1, 0, 1)$ and $A2, B2 = (1, -2, 1)$.

```
#we first give the A's and B's
A1<-c(1,0,-1)
A2<-c(1,-2,1)
B1<-c(1,0,-1)
B2<-c(1,-2,1)

#We create the columns for our design matrix:
A1_col<- rep(A1, each=12)
A2_col<- rep(A2, each=12)
B1_col<- rep(rep(B1,each=4),times=3)
B2_col<- rep(rep(B2,each=4),times=3)
```

Of course for our design matrix we also need to calculate the cross products of those vectors to fill the interaction columns.

```

A1B1<-A1_col*B1_col
A1B2<-A1_col*B2_col
A2B1<-A2_col*B1_col
A2B2<-A2_col*B2_col

```

Having everything ready we create our design matrix.

```

Design <- data.frame(
  Intercept = 1,
  A1 = A1_col,
  A2 = A2_col,
  B1 = B1_col,
  B2 = B2_col,
  A1B1 = A1B1,
  A1B2 = A1B2,
  A2B1 = A2B1,
  A2B2 = A2B2
)
Design

```

	Intercept	A1	A2	B1	B2	A1B1	A1B2	A2B1	A2B2
1	1	1	1	1	1	1	1	1	1
2	1	1	1	1	1	1	1	1	1
3	1	1	1	1	1	1	1	1	1
4	1	1	1	1	1	1	1	1	1
5	1	1	1	0	-2	0	-2	0	-2
6	1	1	1	0	-2	0	-2	0	-2
7	1	1	1	0	-2	0	-2	0	-2
8	1	1	1	0	-2	0	-2	0	-2
9	1	1	1	-1	1	-1	1	-1	1
10	1	1	1	-1	1	-1	1	-1	1
11	1	1	1	-1	1	-1	1	-1	1
12	1	1	1	-1	1	-1	1	-1	1
13	1	0	-2	1	1	0	0	-2	-2
14	1	0	-2	1	1	0	0	-2	-2
15	1	0	-2	1	1	0	0	-2	-2
16	1	0	-2	1	1	0	0	-2	-2
17	1	0	-2	0	-2	0	0	0	4
18	1	0	-2	0	-2	0	0	0	4
19	1	0	-2	0	-2	0	0	0	4
20	1	0	-2	0	-2	0	0	0	4

21	1	0	-2	-1	1	0	0	2	-2
22	1	0	-2	-1	1	0	0	2	-2
23	1	0	-2	-1	1	0	0	2	-2
24	1	0	-2	-1	1	0	0	2	-2
25	1	-1	1	1	1	-1	-1	1	1
26	1	-1	1	1	1	-1	-1	1	1
27	1	-1	1	1	1	-1	-1	1	1
28	1	-1	1	1	1	-1	-1	1	1
29	1	-1	1	0	-2	0	2	0	-2
30	1	-1	1	0	-2	0	2	0	-2
31	1	-1	1	0	-2	0	2	0	-2
32	1	-1	1	0	-2	0	2	0	-2
33	1	-1	1	-1	1	1	-1	-1	1
34	1	-1	1	-1	1	1	-1	-1	1
35	1	-1	1	-1	1	1	-1	-1	1
36	1	-1	1	-1	1	1	-1	-1	1

Having the design matrix we can calculate $\hat{\beta}$.

```
Design <- as.matrix(Design)
Design <- Design[, -9]
Y <- as.matrix(df$Response)

# Response vector
Y <- df$Response

# Coefficients via least squares
beta_hat <- solve(t(Design) %*% Design) %*% t(Design) %*% Y
beta_hat
```

```
      [,1]
Intercept 105.527778
A1        -20.958333
A2         -1.402778
B1         40.333333
B2         -1.027778
A1B1        4.687500
A1B2       11.645833
A2B1       -6.395833
```

We see that the analysis matches the results we expected and the results we derived by doing the calculations by hand. We will now do it with a single line of code to calculate p-values as well.

```
#| echo: true
Design <- data.frame(Design, Response = df$Response)
model_contrasts <- lm(Response ~ A1 + A2 + B1 + B2 + A1B1 + A1B2 + A2B1 + A2B2, data = Design)
summary(model_contrasts)
```

Call:

```
lm(formula = Response ~ A1 + A2 + B1 + B2 + A1B1 + A1B2 + A2B1 +
    A2B2, data = Design)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-60.750	-14.625	1.375	17.938	45.250

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	105.528	4.331	24.367	< 2e-16 ***
A1	-20.958	5.304	-3.951	0.000503 ***
A2	-1.403	3.062	-0.458	0.650565
B1	40.333	5.304	7.604	3.53e-08 ***
B2	-1.028	3.062	-0.336	0.739753
A1B1	4.688	6.496	0.722	0.476759
A1B2	11.646	3.751	3.105	0.004434 **
A2B1	-6.396	3.751	-1.705	0.099622 .
A2B2	2.340	2.165	1.081	0.289364

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

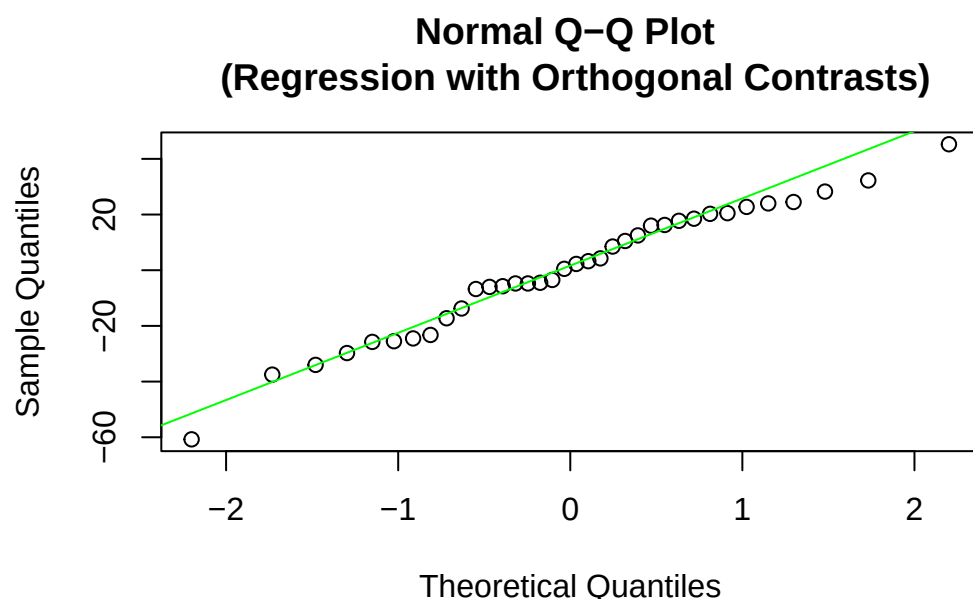
Residual standard error: 25.98 on 27 degrees of freedom

Multiple R-squared: 0.7652, Adjusted R-squared: 0.6956

F-statistic: 11 on 8 and 27 DF, p-value: 9.426e-07

We can also make a Normal Probability Plot with the residuals:

```
qqnorm(residuals(model_contrasts),main = "Normal Q-Q Plot \n (Regression with Orthogonal Contrasts)")
qqline(residuals(model_contrasts),col="green")
```



It's exactly the same as the one for the two-way ANOVA as expected.